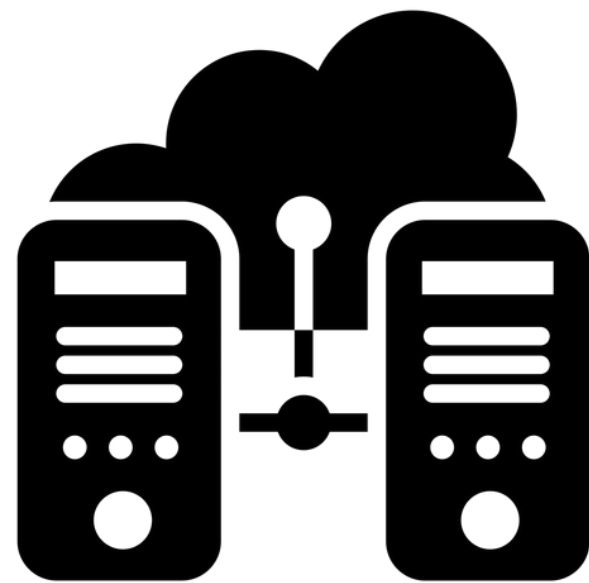




Moving Data Through the Core

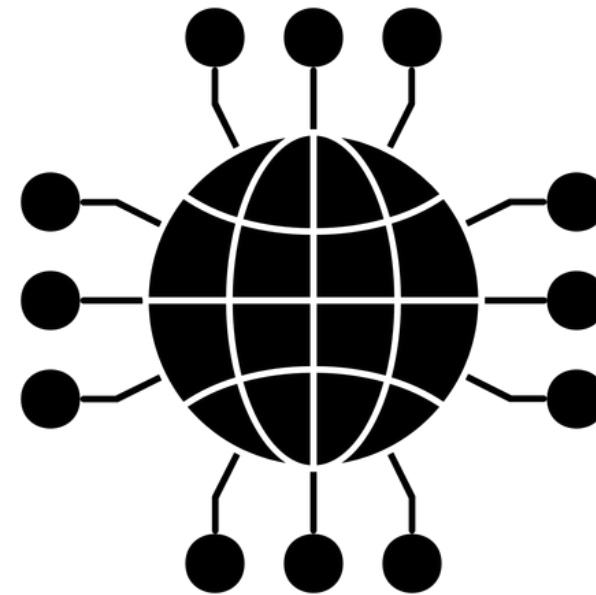
CSAI321:Computer Networks

Recap



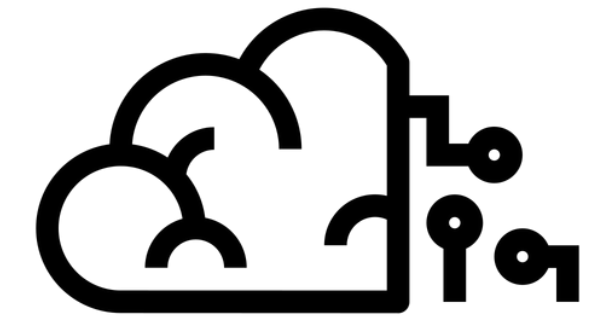
The Physical "Cloud"

The cloud consists of millions of physical servers in large data centers that power our daily applications.



Global Infrastructure

Connecting the world needs significant scale, with over 597 submarine cables linking continents to deliver data rapidly.



The AI Network Era

Modern AI requires immense computing power, utilizing ultra-fast networks to connect over 10,000 GPUs for training advanced models.

Join the lecture online on your dashboard

A Tale of Two Phone Calls



The Landline Call

Grandma pauses for 10 seconds to think? **That capacity sits idle — reserved, but wasted.**



The WhatsApp Call

You pause for 10 seconds? **No packets sent — that capacity is free for someone else.**

The Road Not Taken: Circuit Switching.

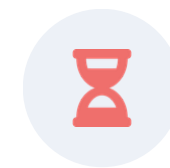


The classic telephone model. Before any conversation could start, the network reserved a dedicated end-to-end path and held it, exclusively, for the full duration of the call.



Guaranteed

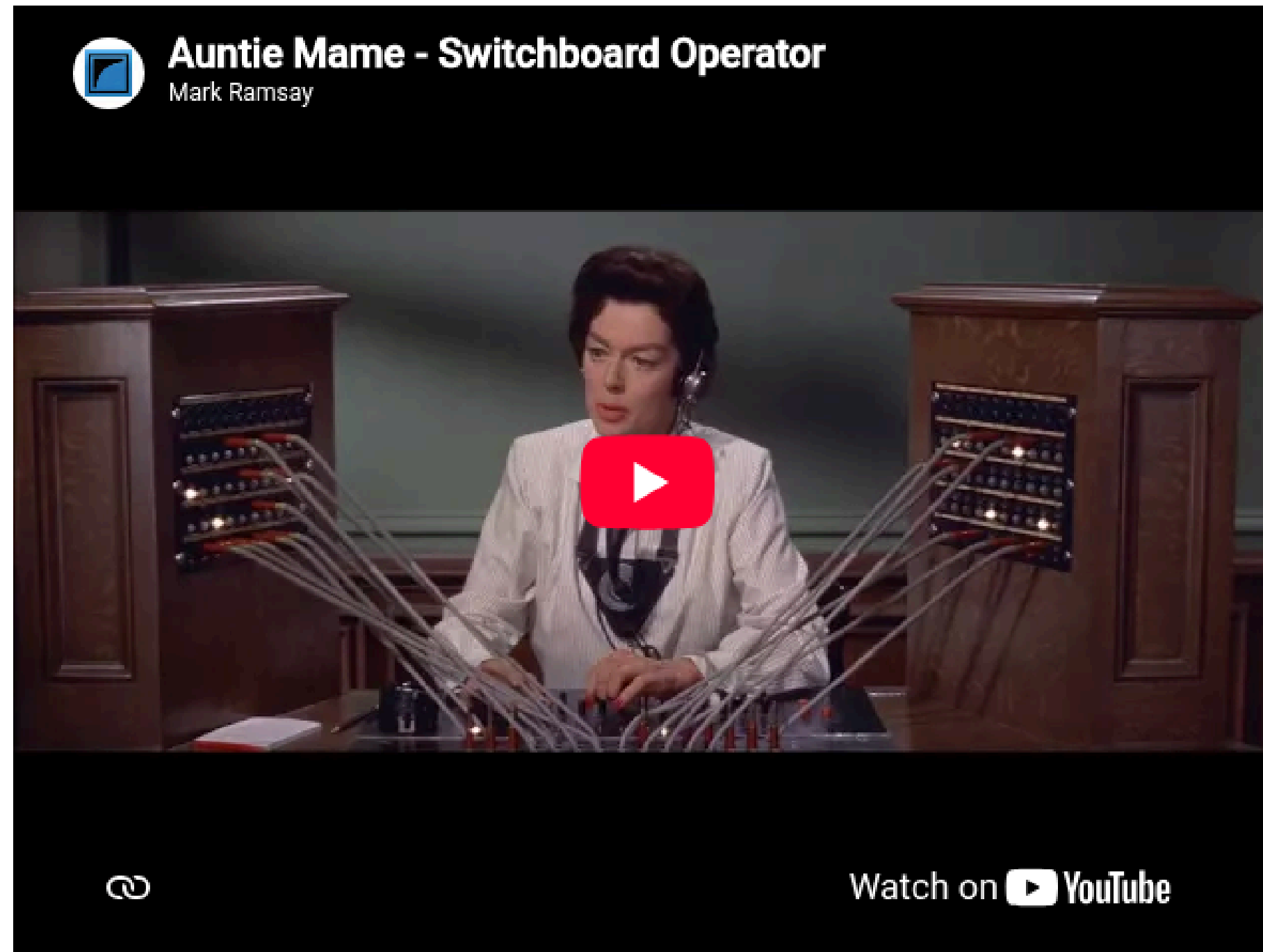
Once the circuit is set up, the full reserved capacity is yours predictable, stable, no surprises mid-call.



Wasteful for Bursty Data

Go quiet for even a few seconds, and that reserved capacity sits idle unusable by anyone else, the whole time.

A Tale of Two Phone Calls



How the Network Core Moves Data

What Exactly Is a “Packet”?



The Idea

Your message — a photo, a webpage, a voice clip — is chopped into small, **independent chunks called packets.**

- Each packet carries a header (addressing info) and a payload (your actual data).
- Packets can take different routes to the same destination.
- The core forwards each packet independently — hop by hop, router to router.



Store-and-Forward Rule

A router must receive the entire packet before it sends even the first bit onward.

PKT



Receive fully → check → forward fully

Why it matters: no single router needs to know the whole path — it just needs to know the next hop.

Sharing on Demand: Statistical Multiplexing.

No reservations. Links are used only when packets actually need them — and that single idea unlocks two superpowers.



Efficiency

Real traffic is bursty — you're rarely sending data every millisecond. Packet switching lets many users share one link, because it's statistically unlikely they all burst at once.

bursty users, one shared link



Resilience

Because packets aren't tied to one fixed path, if a link or router fails, traffic can simply be rerouted around the damage — nothing was “reserved” there to lose.

rerouted around the failure

Packet Switching



Packet switching gives no built-in guarantees. Packets can be delayed behind others in a queue, or dropped entirely when a router's buffer fills up under load.



This Is the Seed of Everything TCP Does

In Weeks 5–6, we'll meet TCP — a protocol whose entire job is to paper over exactly this gap: detecting loss, retransmitting, and pacing data so an unreliable, best-effort core feels reliable to your applications.

Sources Of Delay

The Toll-Booth Caravan

Picture a caravan of cars arriving at a toll booth, then driving down a long highway to the next toll booth.



Each car must individually pass through the booth (*one at a time, takes a fixed time each*), then everyone drives the highway together at the speed limit.

Every Packet Pays Four Kinds of Delay

1



Processing

Router examines the header, checks for bit errors.

tiny, ~microseconds

2



Queuing

Time waiting in a buffer behind other packets.

variable — named only, not computed

3



Transmission

Time to push all the packet's bits onto the link.

$= L / R$ — we compute this

4



Propagation

Time for a bit to physically travel the link.

$= d / s$ — we compute this

Toll Booth = Transmission. Highway = Propagation.



Transmission Delay *“Toll-booth time”*

Time spent pushing every car (every bit) through the booth, one after another, before it can even start down the highway.

Depends on: how many cars (packet size) and how fast the booth processes them (link rate).

MORE BOOTHS → FASTER



Propagation Delay *“highway drive time”*

Time spent actually driving down the highway between the two booths, at a fixed speed limit — has nothing to do with how many cars there are.

Depends on: the distance between booths and the speed limit (signal speed).

MORE LANES ≠ FASTER

Deep Dive: Two Threads Worth Pulling



Why Big Tech Builds Its Own Backbone

Google, Meta, and Netflix lay their own private fibre between data centers and cities — shortening the physical path their traffic must travel.

Connects back to Lecture 1: shorter distance, in d/s terms, means lower propagation delay — the edge/core story and the delay story are the same story.



Why Packet Size Isn't “Bigger Is Better”

The relay demo hinted at this: across many hops, packet size changes how much pipelining is possible.

Huge packets: long waits at each hop before forwarding can start.

Tiny packets: more headers, more per-packet processing overhead.

Let's Do the Maths — Lightly

Transmission Delay: Pushing Bits Onto the Link

$$L / R$$

packet length (bits) $\div$ link rate (bits/sec)



Worked Example

$L = 8,000$ bits on a link with $R = 2$ Mbps

$$8,000 \div 2,000,000 = 4 \text{ ms}$$



Why Doubling the Link Rate Matters

2 Mbps

4 ms

4 Mbps

2 ms

8 Mbps

1 ms

Propagation Delay: The Bit's Physical Journey

$$d / s$$

distance (m) ÷ signal speed ($\approx 2 \times 10^8$ m/s in fibre)

The 500 Mbps Link That Still Feels Slow



Geostationary Satellite Link

It's advertised at a generous **500 Mbps**. But your video call still has that awkward half-second lag where you and the other person keep talking over each other.

The satellite orbits ~35,786 km above the equator. Up to the satellite, then back down — that's roughly 71,500 km the signal must travel, at the speed of light.



↑ 35,786 km ↑

↓ 35,786 km ↓

EARTH

~240 ms round-trip — from physics alone

When Packets Don't Make It

Throughput: What You Actually Get



Throughput

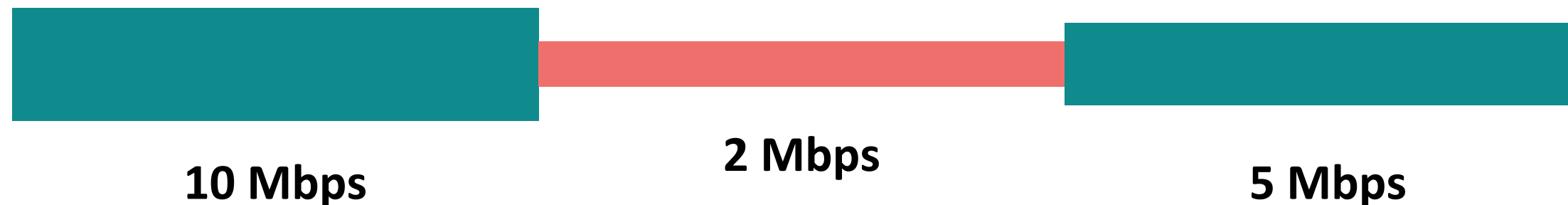
The rate (bits/sec) at which data is actually received — not advertised, not theoretical. What lands.



Bottleneck Link

End-to-end throughput $\approx$ the minimum link rate anywhere on the path. The slowest hop sets the pace for everyone.

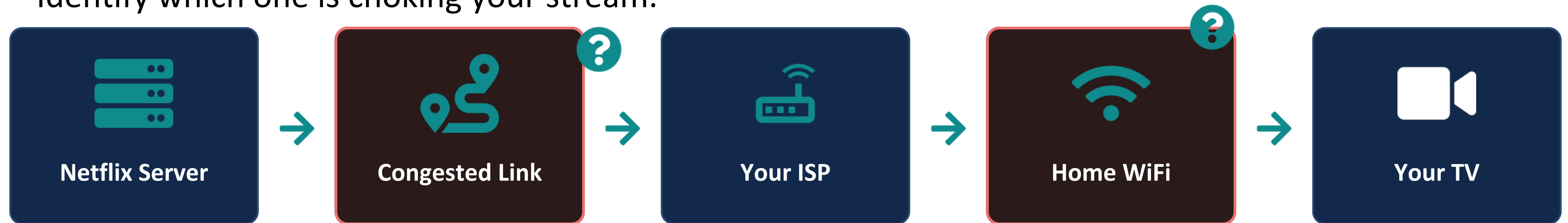
A Path With Three Links



↑ **bottleneck** — caps end-to-end throughput at 2 Mbps, no matter the other links

“I Pay for 300 Mbps. Why Is It Buffering?”

 **Scenario:** Your ISP plan says 300 Mbps. Your 4K stream is stuttering anyway. Where is the real bottleneck hiding? Below is the full path your data travels — each hop has a different capacity. Identify which one is choking your stream.



The 300 Mbps is your access link to the ISP — just one stop on a long path. The real bottleneck could be a congested inter-network link, an overloaded Netflix server, or (very often) your own home WiFi struggling with walls and interference. Next time your stream buffers, think: which hop is the slowest?

The Core Moves Packets. But Who Agrees on the Format?

Thousands of different networks and apps need to agree on a format and order of operations — that's the job of protocol layers.



Lecture 3: OSI and TCP/IP Models

The backbone model for the rest of the course — every later topic (addressing, routing, transport, applications) will be framed in terms of these layers.

NEXT LECTURE

**Thanks
for
watching!**

Please fill the feedback form.